

EG-VQA: Benchmarking Verifiable Video Question Answering with Grounded Temporal Evidence

Linpeng Huang^{1*} Weixing Chen^{1*} Zexin Chen³ Yang Liu^{1†} Liang Lin^{1,2}

¹Sun Yat-sen University ²Peng Cheng Laboratory ³Shenzhen University

*Equal contribution †Corresponding Author

huanglp33@mail2.sysu.edu.cn, chenwx228@mail2.sysu.edu.cn, 2021270164@email.szu.edu.cn

liuy856@mail.sysu.edu.cn, linliang@ieee.org

<https://hclab-sysu.github.io/EG-VQA/>

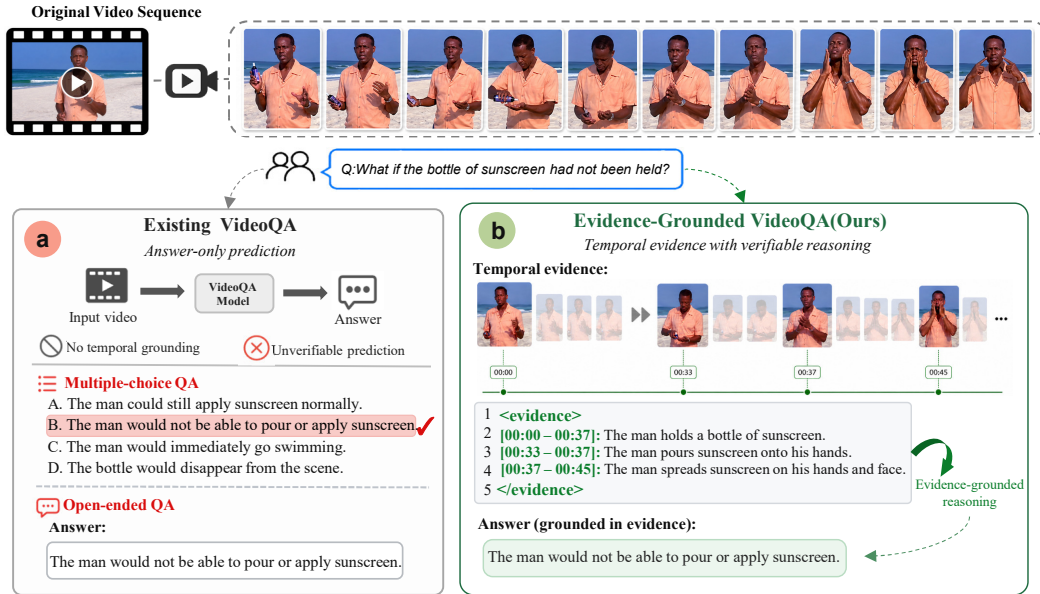


Figure 1: Overview of the proposed evidence-grounded VideoQA paradigm. (a) Existing VideoQA focuses on answer-only prediction, which can lead to correct answers without grounding in relevant video evidence. (b) Our formulation requires models to generate temporally localized evidence alongside answers, enabling verifiable and interpretable reasoning.

Abstract

Recent advances in Video Large Language Models (Video-LLMs) have yielded promising performance on video question answering (VideoQA). Nevertheless, existing benchmarks are predominantly evaluated through answer correctness, while the grounding of predictions in relevant video evidence remains largely unexamined. This disconnect between answer generation and evidence understanding motivates the construction of the **Evidence-Grounded Video Question Answering Benchmark (EG-VQA)**, an open-ended evaluation protocol in which each QA pair is explicitly annotated with supporting temporal evidence, thereby requiring joint reasoning and precise evidence localization. EG-VQA is comprised of 2,067 videos and 11,838 QA pairs with fine-grained evidence annotations. To evaluate predicted evidence, **Evidence-Grounded F1 (EG-F1)** is introduced as a unified metric in which temporal alignment and semantic consistency against ground-truth evidence

are jointly measured. Experimental evaluation reveals that even strong proprietary models struggle to accurately ground their predictions, exposing a fundamental discrepancy between answer correctness and faithful evidence localization. To bridge this gap, **EG-Reasoner**, an evidence-grounded reasoning model trained with explicit supervision, is proposed. State-of-the-art performance is achieved among open-source models, with results competitive against proprietary systems, particularly pronounced gains are observed on reasoning-intensive tasks such as counterfactual questions. These findings demonstrate that scaling alone is insufficient for robust video understanding and that structured evidence supervision is essential for the development of more reliable and interpretable VideoQA systems.

1 Introduction

Video Question Answering (VideoQA) serves as a critical testbed for evaluating a model’s capacity to comprehend dynamic visual content and reason over temporal events [Zhong et al., 2022, Lin and Liu, 2026, Jiang et al., 2025]. Despite recent advances in large multimodal models, a fundamental limitation persists: current benchmarks often fail to distinguish between genuine video understanding and the exploitation of superficial correlations [Ko et al., 2023].

Many widely used VideoQA datasets are constructed in a multiple-choice format, wherein models select answers from predefined options [Tapaswi et al., 2016, Xiao et al., 2021, Wu and Yu, 2024, Fu et al., 2025]. Although this paradigm simplifies evaluation, it inadvertently introduces option bias, allowing models to rely on linguistic shortcuts rather than grounding predictions in visual evidence [Liu et al., 2023b, Chen et al., 2025, Liu et al., 2023a], as shown in Figure 1 (a). Furthermore, fine-grained evidence annotations are typically absent from these benchmarks, rendering it difficult to verify whether relevant video segments support a given prediction. Consequently, models frequently produce black-box answers devoid of interpretable reasoning [Liu et al., 2022]. Even in open-ended settings, evaluation is conventionally restricted to answer correctness, while attention to correct temporal regions is overlooked [Xiao et al., 2024, Lei et al., 2020]. These limitations underscore a critical gap: correct answers do not necessarily imply correct reasoning. For VideoQA systems to be deemed trustworthy, both the content of the answer and its provenance within the video must be explicitly verifiable [Xiao et al., 2024].

Motivated by this challenge, we advocate a paradigm shift in VideoQA—from answer prediction to evidence-grounded reasoning. To this end, we introduce the **Evidence-Grounded Video Question Answering Benchmark (EG-VQA)**, which requires models to produce not only an answer but also temporally localized evidence supporting their prediction, as illustrated in Figure 1(b). This design promotes interpretable, verifiable, and accountable reasoning, addressing the growing demand for explainable multimodal AI. The resulting dataset contains 2,067 videos and 11,838 QA pairs spanning four reasoning categories: descriptive, temporal, causal, and counterfactual.

Extensive evaluation of state-of-the-art multimodal models on this benchmark reveals a striking observation. The models that achieve high answer accuracy frequently fail to identify correct evidence, suggesting that current systems continue to rely on shallow reasoning rather than grounded understanding. This exposes a previously underexplored discrepancy between answer correctness and reasoning faithfulness in VideoQA, indicating that explicit evidence supervision is necessary to move beyond superficial correlations.

To bridge this discrepancy, **EG-Reasoner**, an evidence-aware video reasoning model, is proposed. Trained with explicit evidence supervision, it generates answers jointly with temporally grounded evidence. To overcome the limitations of existing evaluation protocols that treat temporal localization and semantic consistency in isolation, the **Evidence-Grounded F1 (EG-F1)** metric is introduced, which jointly assesses both dimensions between predicted and ground-truth evidence through optimal bipartite matching. Experimental results demonstrate that EG-Reasoner significantly improves both answer correctness and evidence grounding, achieving state-of-the-art performance on EG-VQA. Our contributions are summarized as follows:

- To address the lack of fine-grained, interpretable evidence annotations in existing VideoQA benchmarks, the **EG-VQA** dataset is introduced. Comprising 11,838 open-ended QA

pairs with fine-grained temporal annotations along with textual evidence, it enables joint evaluation of answer correctness and evidence localization.

- To bridge the observed discrepancy between high answer accuracy and poor evidence grounding, **EG-Reasoner** is proposed. Through explicit evidence supervision, this model jointly learns to generate answers and localize supporting temporal segments, achieving state-of-the-art performance among open-source models.
- To overcome the limitations of disjoint temporal and semantic evaluation protocols, we propose the **EG-F1** metric, which jointly assesses temporal alignment and semantic consistency via optimal bipartite matching, thereby providing a faithful measure of reasoning quality.

2 Related Work

2.1 Video Question Answering Benchmarks

Video Question Answering (VideoQA) benchmarks have evolved from short-video understanding toward more complex temporal, causal, and evidence-grounded reasoning. Early datasets such as TGIF-QA [Jang et al., 2017], MovieQA [Tapaswi et al., 2016], and MSRVT-QA [Xu et al., 2017] established VideoQA settings over GIFs, movies, and short web videos, while later benchmarks such as ActivityNet-QA [Yu et al., 2019], TVQA [Lei et al., 2018], NExT-QA [Xiao et al., 2021], and STAR [Wu and Yu, 2024] introduced longer videos, subtitle reasoning, causal reasoning, and situated event understanding. Recent benchmarks such as Video-MME [Fu et al., 2025] further evaluate modern multimodal large language models across diverse video domains and durations, but adopt multiple-choice protocols. Although these datasets have advanced VideoQA evaluation, many of them focus on answer correctness, and multiple-choice settings may introduce option bias, allowing models to exploit correlations among questions and candidate answers without fully grounding their predictions in video content [Chen et al., 2025, Liu et al., 2023b, Wei et al., 2023, Liu et al., 2022]. To verify whether answers are supported by visual evidence, TVQA+ [Lei et al., 2020] extends TVQA with spatio-temporal grounding annotations, and NExT-GQA [Xiao et al., 2024] augments NExT-QA with temporal grounding labels for answer-supporting moments. However, these grounded benchmarks mainly indicate where relevant evidence appears, while providing limited supervision on what semantic evidence within each localized segment supports the answer. To address this gap, we introduce EG-VQA, an evidence-grounded open-ended VideoQA dataset that annotates supporting evidence with both temporal boundaries and textual evidence descriptions.

2.2 Video-LLMs and Evidence-Grounded Reasoning

Recent Video Large Language Models (Video-LLMs) and multimodal large language models have substantially improved video understanding by integrating visual perception, temporal modeling, and language generation. Representative model families such as the Qwen-VL series [Bai et al., 2025b,a], InternVL series [Zhu et al., 2025, Wang et al., 2025a], and VideoChat [Li et al., 2023] demonstrate strong capabilities in video captioning, VideoQA, video dialogue, and long-video understanding. Beyond general video understanding, reasoning-oriented methods further improve temporal and spatio-temporal reasoning: Time-R1 [Wang et al., 2025b] applies reinforcement-learning-based post-training for temporal video grounding, VideoChat-R1 [Li et al., 2025] improves spatio-temporal perception with reinforcement fine-tuning and temporally aware rewards. These works improve answer generation, temporal localization, or reasoning performance, but strong performance on these objectives does not necessarily guarantee faithful evidence grounding. Moreover, answer correctness, temporal alignment, and explanation faithfulness are often evaluated separately. To address this limitation, we train EG-Reasoner to generate temporally localized evidence descriptions, intermediate reasoning, and final answers in a unified evidence-grounded output format.

3 Evidence-Grounded Video Question Answering Benchmark

This section describes the construction pipeline of the Evidence-Grounded Video Question Answering (EG-VQA) Benchmark, including the data curation, reasoning-oriented QA generation with prompt refinement, and two-stage quality control, as shown in Figure 2. More details discussed in A.

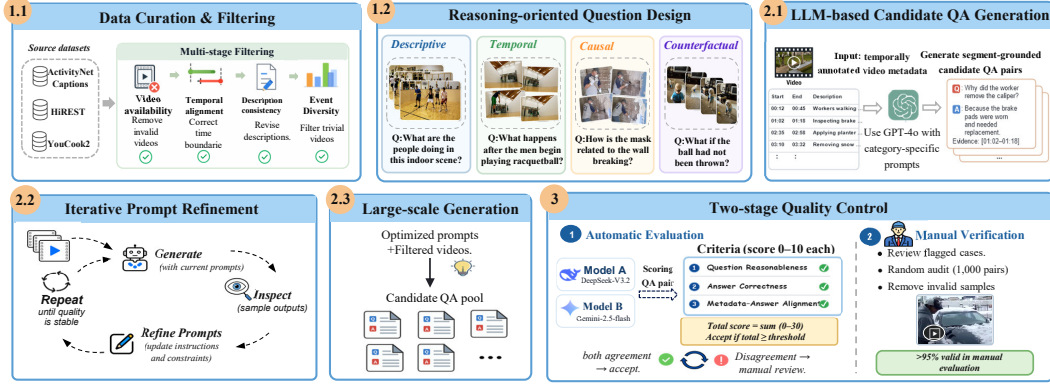


Figure 2: Overview of the EG-VQA dataset construction pipeline.

3.1 Data Curation and Question Design

The foundation of EG-VQA is built from three existing video sources, ActivityNet Captions [Caba Heilbron et al., 2015], HiREST [Zala et al., 2023], and YouCook2 [Zhou et al., 2018], which provide relatively high-quality temporally segmented event annotations across diverse video domains, including open-domain activities, instructional procedures, and long-form event-centric videos. These raw annotations are first standardized into a uniform format where every video is represented as an ordered sequence of timestamped segments paired with concise textual event descriptions. Because the original segment boundaries and descriptions occasionally deviate from the actual video content, a filtering protocol is applied to remove corrupted or inaccessible clips, correct misaligned temporal boundaries, and exclude videos with low event diversity that would otherwise yield trivial questions. The complete filtering criteria are provided in Appendix A.1. After curation, 2,067 high-quality videos remain, with their corrected temporal descriptions serving as the ground-truth evidence backbone.

The questions in EG-VQA target four levels of reasoning. Descriptive(Desc.) questions focus on directly observable content, such as actions, objects, and states. Temporal(Temp.) questions require understanding event order and procedural sequences. Causal(Caus.) questions demand inferring why an event occurs or what consequences it produces. Counterfactual(Cntrf.) questions assess the ability to reason about hypothetical scenarios in which a specific event is altered. These categories ascend in complexity, progressing from surface perception to multi-step reasoning firmly grounded in specific video segments.

3.2 Dataset Construction and Quality Control

QA pairs are generated through a human-in-the-loop pipeline that iterates between automated generation and manual refinement. Given the temporally annotated video metadata, GPT-4o is prompted to produce candidate questions and answers for each reasoning category. The initial prompts are applied to a small held-out subset and manually inspected for common failure modes, including questions answerable without viewing the video, queries weakly tied to the metadata, and answers that omit necessary temporal context. The prompts are refined based on these observations and reapplied until the output consistently satisfies quality requirements. Once finalized, the prompts are deployed at scale across all filtered videos to generate the full candidate pool. Further details on prompt templates and iteration protocols are given in Appendix A.2.

Because large-scale generation can still introduce subtle inconsistencies, a two-stage quality control protocol validates every candidate pair. In the first stage, two independent large language models from distinct architectural families evaluate each sample through a structured scoring rubric that assesses question reasonableness, answer correctness, and the answer-evidence alignment. Samples that pass dual-model agreement are accepted, while disagreements are flagged for manual review. In the second stage, human annotators adjudicate flagged samples and inspect a randomly drawn subset of 1,000 pairs. Over 95% of inspected samples are verified as valid. The full validation rubric, scoring thresholds, and inter-annotator agreement statistics are described in Appendix A.3.

The resulting EG-VQA dataset contains 2,067 videos and 11,838 QA pairs, split in a video-disjoint manner into 8,949 training and 2,889 test samples. Each question is annotated with an average of

2.27 evidence segments, reflecting the multi-step nature of video reasoning. Per-split statistics are reported in Appendix A.4.

3.3 Evaluation via Evidence-Grounded F1

In EG-VQA, a correct response requires both an accurate answer and faithful evidence localization. Existing evaluation protocols typically treat temporal alignment and textual similarity as separate criteria, which allows a model to score well on one dimension while failing on the other. EG-F1 unifies these requirements into a single matching framework in which a prediction is deemed correct only when it aligns with ground-truth evidence in both time and meaning.

Formally, let the ground-truth evidence set be $\mathcal{G} = \{g_i\}_{i=1}^{|\mathcal{G}|}$ and the predicted set be $\mathcal{P} = \{p_j\}_{j=1}^{|\mathcal{P}|}$, where each item $g_i = (t_i, d_i)$ and $p_j = (t_j, d_j)$ consists of a temporal segment and a textual description. The metric proceeds by computing pairwise temporal overlap IoU_{ij} and semantic similarity Sim_{ij} between all pairs, retaining only those edges that satisfy thresholds on both dimensions, and then finding an optimal one-to-one matching via the Hungarian algorithm [Kuhn, 1955]. Precision and recall are derived from the number of matched pairs M , yielding

$$\text{EG-F1} = \frac{2 \cdot (M/|\mathcal{P}|) \cdot (M/|\mathcal{G}|)}{(M/|\mathcal{P}|) + (M/|\mathcal{G}|)}. \quad (1)$$

The complete algorithmic formulation, including threshold settings and the soft variant used during training, is deferred to Appendix B.

4 EG-Reasoner

The training framework of EG-Reasoner, illustrated in Figure 3, casts video question answering as a structured generation task optimized via reinforcement learning. Given an input video and question, the policy model samples multiple candidate responses, and each candidate is scored by a composite reward that measures format compliance, answer correctness, and evidence alignment. Group Relative Policy Optimization Shao et al. [2024], Guo et al. [2025] then compares these candidates to estimate relative quality and update the policy, eliminating the need for a separate critic network.

4.1 Structured Generation Formulation

Given an input pair (V, q) , the model generates a structured response

$$y = (\hat{\mathcal{E}}, \hat{\mathcal{T}}, \hat{\mathcal{A}}), \quad (2)$$

where $\hat{\mathcal{E}} = \{(\hat{t}_k, \hat{d}_k)\}$ is a set of predicted evidence segments with temporal boundaries and textual descriptions, $\hat{\mathcal{T}}$ is a reasoning trajectory that chains the evidence into an explanatory narrative, and $\hat{\mathcal{A}}$ is a concise final answer. This three-block design, shown in the candidate responses of Figure 3, forces the model to commit to both where it looks and how it reasons, making the inference process verifiable rather than opaque.

4.2 Composite Reward Design

The reward function evaluates each candidate response along three complementary dimensions:

$$R(y) = \lambda_f R_{\text{format}}(y) + \lambda_e R_{\text{evi}}(y) + \lambda_a R_{\text{ans}}(y). \quad (3)$$

where the choice of the coefficients $\lambda_f, \lambda_e, \lambda_a$ can be found in the appendix.

Format reward. The model must adhere to a strict output schema containing an `<evidence>` block, a `<think>` block, and an `<answer>` block, as depicted in Figure 3. The format reward is binary, returning 1 if the response follows the required structure and 0 otherwise. This constraint is essential during early training, when the policy might otherwise collapse into unstructured generation.

Answer reward. The generated answer should cover the key elements of the reference while avoiding hallucinated or unsupported content. A rule-guided LLM evaluator judges each response under two principles. The inclusion principle requires the answer to contain as many reference elements as possible; the exclusion principle penalizes content that contradicts the video metadata or cannot be inferred from it. To reduce sparsity in the initial training phase, a relaxed scoring scheme assigns

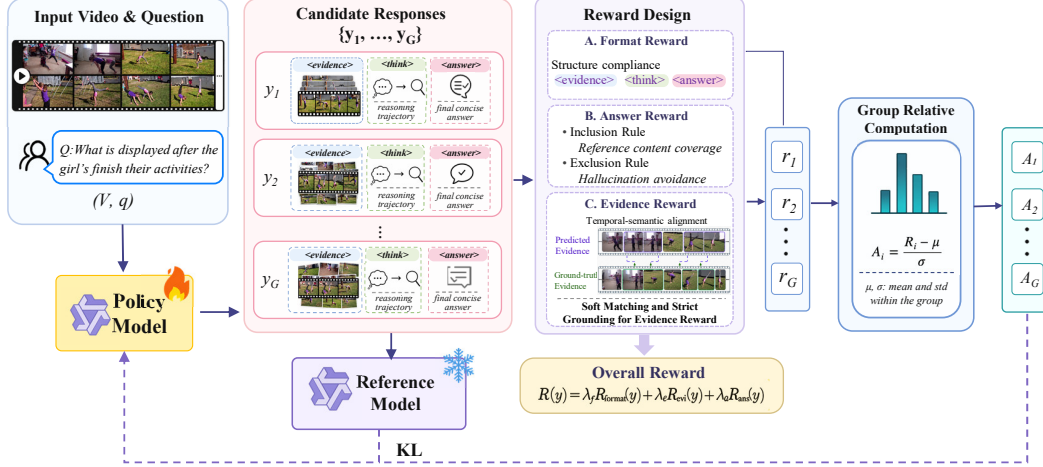


Figure 3: Overview of the EG-Reasoner training framework. Given a video and question, the model generates structured responses containing evidence, reasoning, and answer, and is trained via reinforcement learning with a composite reward that jointly evaluates format, answer correctness, and evidence grounding. GRPO updates the policy by comparing multiple sampled responses.

partial credit to partially correct answers rather than enforcing all-or-nothing judgment. Detailed scoring rules and implementation are provided in Appendix C.1.

Evidence reward. The evidence reward is derived from the EG-F1 metric, but its direct application suffers from sparsity. Because early predictions rarely satisfy the strict temporal and semantic thresholds required for a valid match, the hard thresholding step yields near-zero rewards throughout most training, starving the policy of learning signals for evidence localization.

To mitigate this, a soft matching variant is defined by removing the hard threshold and allowing all predicted-ground-truth pairs to contribute proportionally to their similarity. For a matched pair (i, j) , the contribution is weighted by the product of temporal IoU and semantic similarity, $\text{IoU}_{ij} \cdot \text{Sim}_{ij}$. Soft precision and recall are computed by aggregating these weighted contributions over the optimal bipartite matching, and the soft F1 provides dense gradients even when predictions are imperfect. The complete formulation of the soft variant is provided in Appendix B.2. However, relying solely on soft matching risks tolerates loosely aligned evidence in the final policy. To balance dense supervision with strict correctness, we combine the soft matching score with the original hard EG-F1:

$$R_{\text{evi}} = R_{\text{soft}} + R_{\text{EG-F1}}. \quad (4)$$

This formulation provides dense learning signals during early training while preserving strict alignment constraints in the final policy.

4.3 Group Relative Policy Optimization

The structured output space of EG-Reasoner makes standard supervised fine-tuning insufficient, because a single reference response cannot exhaustively cover all valid evidence orderings or phrasings. Instead, the model is trained with Group Relative Policy Optimization (GRPO), which compares multiple candidates sampled from the same input to estimate relative quality.

As shown in the right half of Figure 3, the current policy π_θ samples a group of G candidate responses $\{y_1, \dots, y_G\}$ for each training instance. Every candidate is scored by the composite reward $R(y_i)$, and its relative advantage is computed by normalizing against the group statistics:

$$A_i = \frac{R_i - \mu}{\sigma}, \quad (5)$$

where μ and σ are the mean and standard deviation of rewards within the group. The policy is then updated to increase the likelihood of high-advantage responses while suppressing low-advantage ones through a clipped importance-weighted objective. A KL divergence penalty from a frozen reference model is added to prevent the policy from departing too far from the initialization. The full objective and variance-reduction analysis are provided in Appendix C.2.

Table 1: Answer performance and evidence grounding performance.

Model	Strict Accuracy					Relaxed Accuracy					Evidence Score							
	Desc.	Temp.	Caus.	Cntrf.	Overall	Desc.	Temp.	Caus.	Cntrf.	Overall	F1@0.1	F1@0.3	F1@0.5	F1@0.7	EG-F1 (0.3,0.5)	EG-F1 (0.3,0.75)	EG-F1 (0.5,0.75)	
Human Results*	45.68	23.87	58.93	46.73	45.44	62.39	51.47	78.56	59.72	65.83	70.02	56.26	21.19	8.43	34.79	6.56	3.87	
Commercial Large Multimodal Models																		
GPT-4o Hurst et al. [2024]	18.9	23.08	37.2	35.1	28.47	40.1	38.94	48.0	53.5	45.05	35.85	17.85	5.67	2.95	8.49	1.18	0.33	
Gemini-2.5-Flash Comanici et al. [2025]	14.58	20.45	33.93	28.89	24.87	37.5	36.36	43.75	46.67	41.19	28.87	19.81	12.56	6.88	12.64	1.48	0.86	
Open-sourced Video-LLMs																		
Qwen2.5-VL-7B Bai et al. [2025b]	9.91	7.73	22.36	16.8	14.31	24.96	20.07	36.21	30.3	27.96	29.36	15.83	7.70	2.55	5.11	0.60	0.35	
InternVL-3-8B Zhu et al. [2025]	9.71	5.24	18.11	13.64	11.71	27.91	18.75	36.45	28.72	26.36	20.89	11.02	4.68	1.34	8.18	1.15	0.66	
Time-R1-7B Wang et al. [2025b]	10.52	7.74	22.73	21.15	15.66	27.52	20.49	39.24	34.24	30.44	46.46	32.03	15.64	5.41	16.46	2.75	1.40	
VideoChat-R1-7B Li et al. [2025]	13.24	9.99	24.1	19.81	16.87	30.29	22.9	41.37	34.15	32.21	48.24	34.50	18.43	6.84	18.71	2.70	1.56	
VideoChat-R1-Thinking-7B Li et al. [2025]	9.91	8.4	24.5	25.65	17.29	26.75	20.97	40.09	38.74	31.76	40.90	26.90	13.69	4.77	14.51	1.82	0.94	
VideoChat-R1.5-7B Yan et al. [2025]	10.65	8.12	22.92	19.02	15.29	28.46	22.04	39.01	32.01	30.42	33.26	21.44	10.88	3.58	11.86	2.17	1.22	
TW-GRPO Dang et al. [2025]	12.37	7.48	25.8	24.21	17.58	27.79	19.56	39.97	36.7	31.14	33.30	19.48	8.97	2.94	9.65	1.07	0.58	
EG-Reasoner (Ours)	14.20	10.92	33.16	48.16	26.88	34.12	27.1	48.46	60.5	42.71	65.97	48.02	27.27	10.30	28.79	4.95	3.12	

Unlike PPO, GRPO eliminates the need for a separate value network by using the group mean as a baseline. This is particularly suitable for EG-Reasoner, where the value function would need to assess the quality of diverse evidence structures, a task that is arguably as difficult as generation itself.

4.4 Training Procedure

The overall training procedure iterates between sampling response groups from the policy, computing composite rewards and relative advantages, and updating the policy parameters. The visual backbone is frozen throughout training; only the language model parameters are optimized. Additional implementation details and hyperparameters are provided in Appendix C.3.

5 Experiments

5.1 Experimental Setup

Models. We evaluate a diverse set of Video-LLMs, including both proprietary and open-source systems. For proprietary models, we consider GPT-4o and Gemini-2.5-Flash. For open-source baselines, we include general-purpose multimodal models (Qwen2.5-VL-7B, InternVL-3-8B) and reasoning-oriented models including Time-R1, TW-GRPO and the VideoChat-R1 series. Detailed descriptions of all baselines are provided in Appendix D.1.

Evaluation Metrics. We evaluate performance from two complementary perspectives: answer correctness and evidence grounding. All tasks in EG-VQA are formulated as open-ended QA. For answer correctness, we adopt an LLM-based evaluation protocol and report both strict and relaxed accuracy. For evidence grounding, we use the proposed EG-F1 metric, together with event-level F1 under multiple temporal IoU thresholds. Detailed evaluation protocols are provided in Appendix D.2 and Appendix D.3, respectively.

Implementation details. We adopt Qwen2.5-VL-7B-Instruct as the base model of EG-Reasoner and fine-tune it on the EG-VQA training split. During both training and evaluation, the model is required to generate structured outputs following the unified format defined in Section 4. Detailed training hyperparameters and evaluation configurations are provided in Appendix C.3 and Appendix D.4.

Human study. To set a human upper bound and validate our evaluation protocol, we conduct a human study on a subset of the EG-VQA test set. We randomly sample 100 QA pairs covering all question categories and ask three human annotators to provide open-ended answers with supporting evidence under the same setting as models. The collected responses are evaluated using the same answer correctness and evidence grounding metrics. We report the average performance across annotators.

5.2 Main Results

Table 1 summarizes the performance of all models on EG-VQA. Overall, existing Video-LLMs struggle on the benchmark, with most open-source baselines achieving less than 20% strict accuracy. Even strong proprietary models remain far from saturated performance, indicating that EG-VQA poses substantial challenges for both open-ended reasoning and evidence grounding.

Among all open-source models, **EG-Reasoner** achieves the best overall performance, reaching **26.88%** strict accuracy and **42.71%** relaxed accuracy, representing a **+12.57%** improvement over the base model Qwen2.5-VL-7B. Notably, EG-Reasoner performs competitively with proprietary systems and surpasses GPT-4o on counterfactual questions (**48.16%** vs. **35.10%**). In addition to answer correctness, EG-Reasoner achieves the strongest evidence grounding performance among all

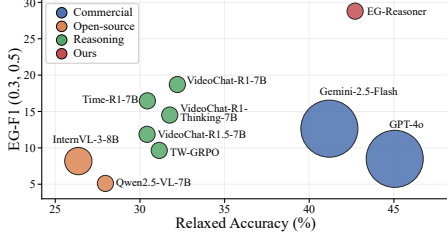


Figure 4: Comparison between answer correctness and evidence grounding.

Model	EG-VQA			
	Strict Acc.↑	Relaxed Acc.↑	Avg. F1↑	Avg. EG-F1↑
EG-Reasoner (Ours)	26.88	42.71	37.89	12.28
w/o Evidence Reward	22.21	36.17	15.81	4.21
w/o Soft Evidence Reward	24.00	37.96	15.33	3.62
VideoChat-R1-7B Li et al. [2025]	16.78	32.21	27.00	7.65
+RL w/o Evidence Reward	25.42	40.26	22.07	6.51

Table 2: Ablation results on answer accuracy and evidence grounding.

evaluated models, including proprietary large models (GPT-4o and Gemini-2.5-flash). In particular, it reaches **27.27 F1@0.5 (+19.57 over the base model)**, with consistent gains across all EG-F1 settings.

Human performance remains substantially higher than all existing models on overall answer correctness, indicating that EG-VQA remains highly challenging even for state-of-the-art Video-LLMs. Interestingly, EG-Reasoner surpasses average human performance on several evidence localization metrics, including counterfactual relaxed accuracy, F1@0.5, and F1@0.7. We attribute this to the fact that human annotators typically provide concise evidence annotations, whereas EG-Reasoner generates denser evidence predictions that better align with the evaluation protocol. Nevertheless, humans still maintain stronger overall reasoning and semantic understanding.

5.3 Analysis and Discussion

Answer correctness does not imply correct grounding. Figure 4 visualizes the relationship between answer accuracy and evidence grounding across models. A clear discrepancy can be observed: models with strong answer accuracy do not necessarily achieve high grounding performance. For example, proprietary models obtain competitive relaxed accuracy, but their evidence grounding remains limited. This indicates that current Video-LLMs can often produce plausible answers without accurately identifying the supporting temporal evidence. In contrast, EG-Reasoner improves both dimensions, suggesting that explicit evidence supervision helps align answers with video evidence.

Precise temporal localization remains challenging. As shown in Table 1, all models show performance degradation as the IoU threshold increases from 0.1 to 0.7, indicating that precise temporal boundary localization remains challenging. Nevertheless, EG-Reasoner consistently outperforms all baselines across the full IoU range, demonstrating stronger fine-grained temporal grounding ability.

Evidence grounding particularly benefits complex reasoning. Table 1 shows that the performance gains of EG-Reasoner are pronounced on causal and counterfactual questions, which require reasoning over event dependencies and hypothetical scenarios. In particular, EG-Reasoner improves counterfactual strict accuracy from 16.80% to 48.16% compared with the base model. These results suggest that explicit evidence supervision is particularly beneficial for complex multi-step reasoning.

Scaling alone is insufficient. The marker size in Figure 4 reflects model scale. Larger proprietary models occupy the high-answer but low-grounding region, whereas EG-Reasoner, despite being smaller, moves closer to the upper-right region with stronger evidence grounding. This suggests that scaling alone does not resolve grounding failures; explicit supervision for evidence localization is necessary for faithful and verifiable video reasoning.

5.4 Ablation Studies

Effect of evidence supervision. We first examine the role of evidence-grounded supervision. Removing the evidence reward leads to consistent performance drops in both answer correctness and evidence grounding. As shown in Table 2, the overall relaxed accuracy drops from 42.71% to 37.96%, while the evidence grounding score (EG-F1) also decreases significantly. This shows that explicit supervision is essential for learning accurate evidence localization. To further verify that the improvement is not due to model choice, we select VideoChat-R1-7B as a representative strong baseline, which achieves competitive performance among open-source models on EG-VQA (32.21% relaxed accuracy). We then train this model using the same reinforcement learning framework but without evidence reward. This variant still underperforms EG-Reasoner across all metrics. These results show that exposure to training data alone is insufficient and that explicit evidence-grounded supervision is the key factor driving performance gains.

Effect of soft evidence design. We evaluate the impact of the proposed soft evidence reward by replacing it with the original hard EG-F1. This leads to significant performance drops, particularly in evidence grounding metrics (e.g., Avg. EG-F1 drops from 12.28 to 3.62), showing a failure to learn accurate temporal localization. We attribute this to the sparsity of hard threshold-based reward, which provides limited learning signals during training. Interestingly, replacing the soft evidence reward with hard EG-F1 leads to even worse grounding performance than removing the evidence reward entirely. This surprising finding suggests that sparse and unstable reward signals can be more harmful than having no evidence reward at all, because they provide misleading or ineffective optimization feedback during training. The soft evidence design gives partial credit to near-matched evidence, providing denser supervision and more stable optimization, which leads to improved performance.

5.5 Case Study

Figure 5 presents qualitative comparisons between the base model Qwen2.5-VL and EG-Reasoner in evidence-grounded reasoning. In Figure 5(a), the baseline model predicts the correct answer but fails to localize the relevant temporal segment, instead relying on loosely related context. In contrast, EG-Reasoner accurately identifies the key evidence region associated with the chalk-drawing action. Figure 5(b) further demonstrates the advantage of evidence-grounded reasoning in complex procedural understanding. The baseline model produces hallucinated objects and incorrect reasoning steps, whereas EG-Reasoner decomposes the task into fine-grained temporal evidence segments and generates a coherent reasoning chain aligned with the video content, leading to the correct answer. Additional qualitative examples and failure cases are provided in Appendix E.

a

Video:

Question: What does the man do with the chalk during the activity?

GT Evidence: 01:25 01:37

GT Answer: The man draws on the concrete with chalk. *Draw on the concrete with chalk*

Baseline (Qwen2.5-VL)	EG-Reasoner (Ours)
<i><evidence></i> 00:25-00:30, The man explains the rules of the game to the group. ✗ <i><evidence></i> 00:30-01:30, The man and the group start playing hopscotch. ✗ <i><evidence></i> The man uses chalk to draw hopscotch squares. ✓	<i><evidence></i> 01:20-01:40, The man is using chalk to draw hopscotch boxes on the ground. ✓ <i><evidence></i> The man uses the chalk to draw hopscotch boxes on the ground. ✓

b

Video:

Question: What are the steps taken by the person to carve the pumpkin?

GT Evidence: 00:00 00:10 00:11 00:35 00:35 01:23

GT Answer: *cutting a pumpkin. out the pumpkin and traces the picture outline. Cuts along the outline and shows the final result.*
The person cuts open the pumpkin, scoops it out, traces a picture, then cuts the outline.

Baseline (Qwen2.5-VL)	EG-Reasoner (Ours)
<i><evidence></i> 00:00-00:05, ... removes the protector from a pumpkin. ✗ <i><evidence></i> 00:05-00:20, ... cuts the top of the pumpkin with a knife. ✗ <i><evidence></i> 00:20-00:40, ... places a canyon 8 sticker on the pumpkin. ✗ <i><evidence></i> 00:40-00:55, ... using a plastic Department 56 3D tool. ✗ <i><evidence></i> 01:05-01:15, ... stops carving and places a small sharpening tool, a file, and a light on the table. ✗ <i><evidence></i> The person steps to carve the pumpkin by removing the protector, cutting the top, placing a sticker, and then carving the pumpkin using a Department 56 3D tool. ✗	<i><evidence></i> 00:00-00:15, ... slices the pumpkin open. ✓ <i><evidence></i> 00:15-00:25, ... removes the pumpkin's insides with a scoop. ✓ <i><evidence></i> 00:25-00:35, ... positions a picture on the pumpkin and marks its outline. ✓ <i><evidence></i> 00:35-01:18, ... uses a knife to carve the pumpkin along the traced lines. ✓ <i><evidence></i> The person slices the pumpkin open and hollows it out, outlines a picture on it, and then carves along the traced lines. ✓

Figure 5: Demonstrations of Baseline Models and Our EG-Reasoner. Hallucinated evidence is highlighted in red, and correctly grounded evidence is marked in green.

6 Conclusion

This paper introduces EG-VQA, a benchmark that requires models to ground their answers in explicit temporal evidence. Additionally, the Evidence-Grounded F1 (EG-F1) metric is proposed to jointly measure temporal alignment and semantic consistency of evidence. Experiments show that current Video-LLMs, including strong proprietary models, struggle to consistently ground their predictions,

highlighting a fundamental gap between answer correctness and evidence localization. To address this issue, we propose EG-Reasoner, an evidence-grounded reasoning model trained with explicit supervision, which achieves state-of-the-art performance among open-source models and competitive results compared to proprietary systems. Overall, our findings suggest that incorporating structured evidence supervision is essential for reliable and interpretable video understanding, and we hope EG-VQA can serve as a useful testbed for future research on evidence-aware multimodal reasoning.

References

- Shuai Bai, Yuxuan Cai, Ruizhe Chen, Keqin Chen, Xionghui Chen, Zesen Cheng, Lianghao Deng, Wei Ding, Chang Gao, Chunjiang Ge, et al. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025a.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025b.
- Fabian Caba Heilbron, Victor Escorcia, Bernard Ghanem, and Juan Carlos Niebles. Activitynet: A large-scale video benchmark for human activity understanding. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 961–970, 2015.
- Weixing Chen, Yang Liu, Binglin Chen, Jiandong Su, Yongsun Zheng, and Liang Lin. Cross-modal causal relation alignment for video question grounding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 24087–24096, 2025.
- Seongho Choi, Kyoung-Woon On, Yu-Jung Heo, Ahjeong Seo, Youwon Jang, Minsu Lee, and Byoung-Tak Zhang. Dramaqa: Character-centered video story understanding with hierarchical qa. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 1166–1174, 2021.
- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Noveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- Jisheng Dang, Jingze Wu, Teng Wang, Xuanhui Lin, Nannan Zhu, Hongbo Chen, Wei-Shi Zheng, Meng Wang, and Tat-Seng Chua. Reinforcing video reasoning with focused thinking. *arXiv preprint arXiv:2505.24718*, 2025.
- Chaoyou Fu, Yuhao Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24108–24118, 2025.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*, 2024.
- Zaeem Hussain, Mingda Zhang, Xiaozhong Zhang, Keren Ye, Christopher Thomas, Zuha Agha, Nathan Ong, and Adriana Kovashka. Automatic understanding of image and video advertisements. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1705–1715, 2017.
- Yunseok Jang, Yale Song, Youngjae Yu, Youngjin Kim, and Gunhee Kim. Tgif-qa: Toward spatio-temporal reasoning in visual question answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2758–2766, 2017.
- Kaixuan Jiang, Yang Liu, Weixing Chen, Jingzhou Luo, Ziliang Chen, Ling Pan, Guanbin Li, and Liang Lin. Beyond the destination: A novel benchmark for exploration-aware embodied question answering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9091–9101, 2025.
- Dohwan Ko, Ji Lee, Woo-Young Kang, Byungseok Roh, and Hyunwoo Kim. Large language models are temporal and causal reasoners for video question answering. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 4300–4316, 2023.

- Harold W Kuhn. The hungarian method for the assignment problem. *Naval research logistics quarterly*, 2(1-2):83–97, 1955.
- Jie Lei, Licheng Yu, Mohit Bansal, and Tamara Berg. Tvqa: Localized, compositional video question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing*, pages 1369–1379, 2018.
- Jie Lei, Licheng Yu, Tamara Berg, and Mohit Bansal. Tvqa+: Spatio-temporal grounding for video question answering. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 8211–8225, 2020.
- Kunchang Li, Yinan He, Yi Wang, Yizhuo Li, Wen Wang, Ping Luo, Yali Wang, Limin Wang, and Yu Qiao. Videochat: chat-centric video understanding. *Science China Information Sciences*, 68, 2023. URL <https://api.semanticscholar.org/CorpusID:258588306>.
- Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, et al. Mvbench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 22195–22206, 2024.
- Linjie Li, Yen-Chun Chen, Yu Cheng, Zhe Gan, Licheng Yu, and Jingjing Liu. Hero: Hierarchical encoder for video+ language omni-representation pre-training. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 2046–2065, 2020.
- Xinhao Li, Ziang Yan, Desen Meng, Lu Dong, Xiangyu Zeng, Yinan He, Yali Wang, Yu Qiao, Yi Wang, and Limin Wang. Videochat-r1: Enhancing spatio-temporal perception via reinforcement fine-tuning. *arXiv preprint arXiv:2504.06958*, 2025.
- Liang Lin and Yang Liu. Multimodal large models: A new paradigm of artificial intelligence. *Cham: Springer*, 2026.
- Y. Liu, Weixing Chen, Guanbin Li, and Liang Lin. Causalv1r: A toolbox and benchmark for visual-linguistic causal reasoning. *ArXiv*, abs/2306.17462, 2023a. URL <https://api.semanticscholar.org/CorpusID:259308889>.
- Yang Liu, Yu-Shen Wei, Hong Yan, Guan-Bin Li, and Liang Lin. Causal reasoning meets visual representation learning: A prospective study. *Machine Intelligence Research*, 19(6):485–511, 2022.
- Yang Liu, Guanbin Li, and Liang Lin. Cross-modal causal relational reasoning for event-level visual question answering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10): 11624–11641, 2023b.
- Xinwei Long, Kai Tian, Peng Xu, Guoli Jia, Jingxuan Li, Sa Yang, Yihua Shao, Kaiyan Zhang, Che Jiang, Hao Xu, et al. Adsqa: Towards advertisement video understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 23396–23407, 2025.
- Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *Advances in Neural Information Processing Systems*, 36:46212–46244, 2023.
- Munan Ning, Bin Zhu, Yujia Xie, Bin Lin, Jiayi Cui, Lu Yuan, Dongdong Chen, and Li Yuan. Video-bench: A comprehensive benchmark and toolkit for evaluating video-based large language models. *Computational Visual Media*, 2025.
- Ruchit Rawal, Khalid Saifullah, Miquel Farré, Ronen Basri, David Jacobs, Gowthami Somepalli, and Tom Goldstein. Cinepile: A long video question answering dataset and benchmark. *arXiv preprint arXiv:2405.08813*, 2024.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

- Makarand Tapaswi, Yukun Zhu, Rainer Stiefelhof, Antonio Torralba, Raquel Urtasun, and Sanja Fidler. Movieqa: Understanding stories in movies through question-answering. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4631–4640, 2016.
- Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. Internv13. 5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025a.
- Ye Wang, Ziheng Wang, Boshen Xu, Yang Du, Kejun Lin, Zihan Xiao, Zihao Yue, Jianzhong Ju, Liang Zhang, Dingyi Yang, et al. Time-r1: Post-training large vision language model for temporal video grounding. *arXiv preprint arXiv:2503.13377*, 2025b.
- Yushen Wei, Yang Liu, Hong Yan, Guanbin Li, and Liang Lin. Visual causal scene refinement for video question answering. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 377–386, 2023.
- Bo Wu and Shoubin Yu. Star: A benchmark for situated reasoning in real-world videos. *ArXiv*, abs/2405.09711, 2024. URL <https://api.semanticscholar.org/CorpusID:244907062>.
- Junbin Xiao, Xindi Shang, Angela Yao, and Tat seng Chua. Next-qa: Next phase of question-answering to explaining temporal actions. *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9772–9781, 2021. URL <https://api.semanticscholar.org/CorpusID:234763093>.
- Junbin Xiao, Angela Yao, Yicong Li, and Tat-Seng Chua. Can i trust your answer? visually grounded video question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13204–13214, 2024.
- Dejing Xu, Zhou Zhao, Jun Xiao, Fei Wu, Hanwang Zhang, Xiangnan He, and Yueting Zhuang. Video question answering via gradually refined attention over appearance and motion. In *Proceedings of the 25th ACM international conference on Multimedia*, pages 1645–1653, 2017.
- Ziang Yan, Xinhao Li, Yinan He, Zhengrong Yue, Xiangyu Zeng, Yali Wang, Yu Qiao, Limin Wang, and Yi Wang. Videochat-r1. 5: Visual test-time scaling to reinforce multimodal reasoning by iterative perception. *arXiv preprint arXiv:2509.21100*, 2025.
- Zhou Yu, Dejing Xu, Jun Yu, Ting Yu, Zhou Zhao, Yueting Zhuang, and Dacheng Tao. Activitynet-qa: A dataset for understanding complex web videos via question answering. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pages 9127–9134, 2019.
- Abhay Zala, Jaemin Cho, Satwik Kottur, Xilun Chen, Barlas Oguz, Yashar Mehdad, and Mohit Bansal. Hierarchical video-moment retrieval and step-captioning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23056–23065, 2023.
- Hongjie Zhang, Lu Dong, Yi Liu, Yifei Huang, Yali Wang, Limin Wang, and Yu Qiao. Lvbench: A benchmark for long-form video understanding with versatile multi-modal question answering. *International Journal of Computer Vision*, 133(11):7726–7747, 2025.
- Yaoyao Zhong, Wei Ji, Junbin Xiao, Yicong Li, Weihong Deng, and Tat-Seng Chua. Video question answering: Datasets, algorithms and challenges. In *Proceedings of the 2022 conference on empirical methods in natural language processing*, pages 6439–6455, 2022.
- Luowei Zhou, Chenliang Xu, and Jason Corso. Towards automatic learning of procedures from web instructional videos. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.
- Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. Internv13: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*, 2025.

A Dataset Construction Details

A.1 Filtering protocol

To ensure the reliability and consistency of the benchmark, we apply a structured filtering protocol to the collected videos and their associated annotations. This process is designed to correct annotation errors, remove low-quality samples, and retain videos that support meaningful reasoning.

Unified Metadata Format. The source datasets used in EG-VQA adopt different annotation schemas and temporal segmentation formats. To facilitate unified processing and evidence-grounded question generation, all videos are first converted into a standardized metadata representation consisting of video-level metadata, temporally ordered timestamps, and corresponding event descriptions.

Each video is represented as a structured sequence of timestamp–description pairs. An example is shown below:

```
"TEOPA76qOKA": {
  "duration": 88,
  "title": "How to Make a Lemon Battery",
  "timestamps": [
    [
      30.0,
      34.0
    ],
    [
      34.0,
      53.0
    ],
    [
      53.0,
      63.0
    ],
    [
      63.0,
      76.0
    ],
    [
      76.0,
      85.0
    ]
  ],
  "descriptions": [
    "Roll the lemons",
    "Put copper wire and paper clips",
    "Connect alligator clips",
    "Touch the unconnected ends",
    "Connect the positive to LED"
  ]
}
```

This unified representation serves as the basis for subsequent filtering, question generation, evidence annotation, and quality control procedures.

Data Validation. We first remove corrupted, incomplete, or inaccessible video samples. Videos that cannot be decoded, contain missing segments, or fail to load correctly are excluded from further processing.

Temporal Alignment Refinement. Although the source datasets provide temporally segmented annotations, the original segment boundaries and descriptions may not always accurately reflect the underlying video content. To address this, human annotators are instructed to review each video and

its corresponding segments, and to refine temporal boundaries when necessary. Segments that exhibit significant misalignment with the actual visual events are corrected or removed.

Description Consistency Check. We further examine the consistency between segment descriptions and the observed video content. Descriptions that are ambiguous, incomplete, or inconsistent with the visual evidence are revised or discarded. Samples with severely noisy or unreliable metadata are excluded entirely.

Informational Density Filtering. To ensure that the resulting dataset supports non-trivial reasoning, we remove videos with low informational density. Specifically, videos that contain only a single event, minimal temporal variation, or insufficient interaction between entities are filtered out, as they tend to yield trivial or purely descriptive questions.

Human Verification. All refinement steps are conducted by trained human annotators following a standardized review protocol. Annotators are instructed to prioritize both temporal accuracy and semantic consistency when validating segment annotations.

After applying the above filtering criteria, we retain a total of 2,067 high-quality videos, which serve as the foundation for constructing the EG-VQA benchmark.

A.2 Prompt Design and Iterative Refinement

We construct question–answer (QA) pairs using a human-in-the-loop pipeline that combines large language model generation with iterative prompt refinement. The goal is to produce high-quality, reasoning-oriented questions that are grounded in temporally annotated video content.

Initial Prompt Design. Given the temporally segmented video metadata, we design type-specific prompts for each reasoning category (descriptive, temporal, causal, and counterfactual). Each prompt explicitly encourages the generation of open-ended questions that require reasoning over one or multiple temporal segments, rather than relying on superficial patterns or static descriptions.

Failure Mode Identification. Direct application of large language models often results in suboptimal QA pairs. To systematically identify common issues, we apply the initial prompts to a held-out subset of videos and manually inspect the outputs. The most frequent failure modes include:

- **Video-independent questions:** Questions that can be answered without referring to the video content.
- **Weak grounding:** Questions or answers that are only loosely related to the temporal metadata.
- **Incomplete reasoning:** Answers that omit necessary temporal or contextual information.

Iterative Refinement. Based on these observations, we iteratively refine the prompts by introducing stricter constraints on grounding, temporal specificity, and reasoning depth. This process follows a generate–inspect–refine loop, where prompts are repeatedly adjusted and re-evaluated until the generated QA pairs consistently satisfy our quality criteria. During this process, we explicitly filter templated or weakly grounded questions to reduce potential shortcut patterns that could be exploited without genuine video understanding.

Final Deployment. Once stabilized, the refined prompts are applied to the full set of filtered videos to generate candidate QA pairs at scale. This iterative design ensures that the resulting dataset emphasizes non-trivial, reasoning-intensive, and evidence-grounded questions.

A.3 Quality Control Protocol

To ensure the reliability of generated QA pairs, we employ a two-stage quality control protocol that combines structured automatic evaluation with human verification.

Stage 1: Structured Multi-model Evaluation. Each candidate QA pair is independently evaluated by two large language models from distinct architectural families (DeepSeek-V3.2 and Gemini-2.5-Flash). We adopt a structured scoring rubric consisting of three criteria:

- **Question Reasonableness:** whether the question is well-posed and requires video understanding.
- **Answer Correctness:** whether the answer is consistent with the video content.
- **Evidence Alignment:** whether the answer is properly grounded in the relevant temporal segments.

Each dimension is scored on a scale of 0–10, and the total score is obtained by summing the three dimensions. A sample is considered acceptable if the total score is greater than or equal to 25.

Agreement-based Filtering. We adopt a dual-model agreement strategy to improve robustness. When both models agree on the decision (accept or reject), the result is directly adopted. Otherwise, the sample is flagged for manual review.

Stage 2: Human Adjudication. All flagged samples are reviewed by human annotators, who determine their validity based on the same evaluation criteria. Annotators are instructed to prioritize both semantic correctness and grounding fidelity.

Post-hoc Validation. After dataset construction, we perform additional manual inspection on a randomly sampled subset of 1,000 QA pairs. More than 95% of the sampled pairs are verified as valid, demonstrating the effectiveness of the quality control protocol.

Inter-annotator Consistency. To ensure consistency, disagreements among annotators are resolved through discussion, and annotation guidelines are refined accordingly.

A.4 Dataset Statistics and Benchmark Comparison

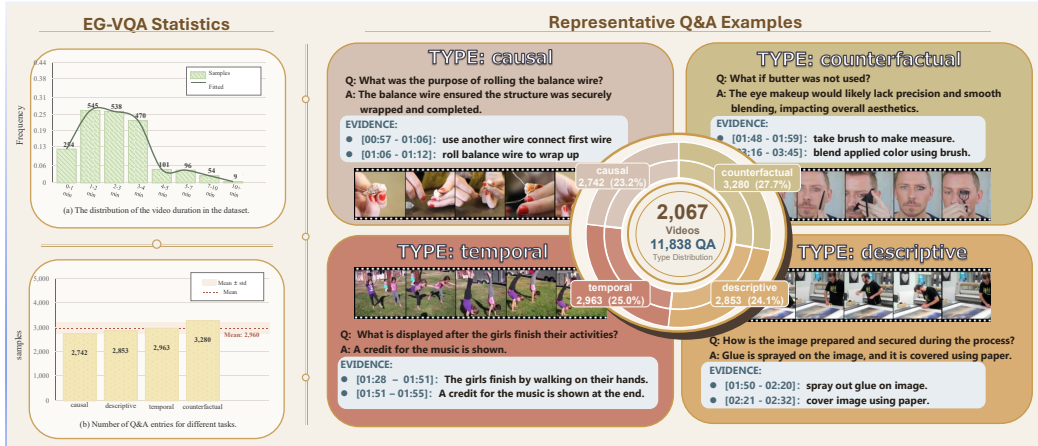


Figure 6: Overview of EG-VQA statistics and representative QA examples.

As shown in Figure 6, the EG-VQA benchmark contains 2,067 videos and 11,838 QA pairs. The dataset is split in a video-disjoint manner into 8,949 training samples and 2,889 test samples to prevent overlap between training and evaluation videos. Detailed per-split statistics are summarized in Table 3.

Reasoning-oriented Question Distribution. EG-VQA contains four categories of reasoning tasks: descriptive, temporal, causal, and counterfactual. These categories progressively increase the reasoning difficulty, ranging from direct perception of observable content to hypothetical reasoning over

altered events. As shown in Figure 6, the dataset maintains a relatively balanced distribution across different reasoning types, ensuring diverse evaluation of video understanding capability.

Table 3: Per-split statistics of EG-VQA.

Question Type	Train		Test	
	#Pairs	Avg. Evid.	#Pairs	Avg. Evid.
Descriptive	2196	2.12	657	2.24
Temporal	2212	2.66	751	2.89
Causal	1994	2.23	748	2.26
Counterfactual	2547	1.96	733	2.04
Total	8,949	2.24	2,889	2.36

Evidence-grounded Reasoning Characteristics. Unlike conventional VideoQA benchmarks that evaluate only answer correctness, EG-VQA explicitly associates each question with temporally localized evidence segments. On average, each question is paired with 2.27 evidence segments, reflecting the multi-step nature of video reasoning.

Figure 7(a) shows the distribution of evidence counts per question. Most questions require multiple evidence segments rather than a single localized event, indicating that successful reasoning often depends on integrating information across temporally separated video events.

In addition, the temporal spans of evidence vary substantially across samples. Figure 7(b) illustrates the distribution of evidence duration. The benchmark contains both short fine-grained actions and long procedural activities, requiring models to perform reasoning at multiple temporal scales.

The average video duration is 177.6 seconds, while the average evidence segment length is 34.4 seconds. This large gap further increases the difficulty of precise temporal grounding, since models must identify a relatively small set of relevant moments within long video contexts.

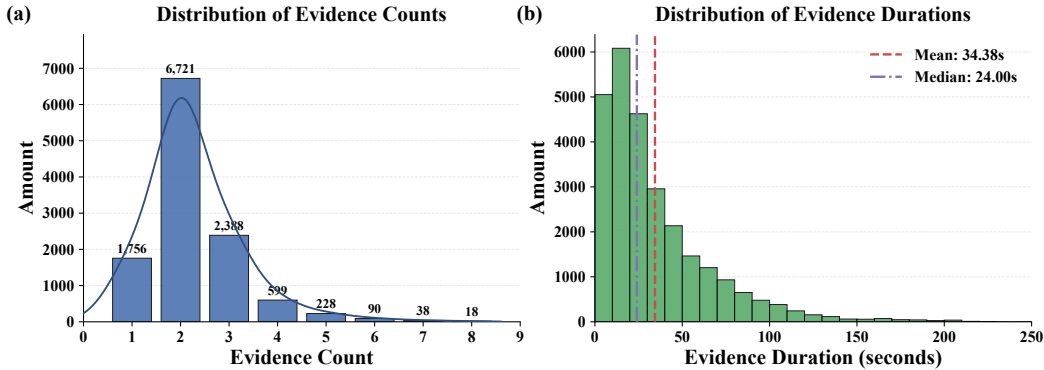


Figure 7: Evidence statistics of EG-VQA. (a) Distribution of evidence counts per question. (b) Distribution of evidence duration.

Benchmark Comparison. Table 4 compares EG-VQA with representative existing VideoQA benchmarks. Most prior datasets either adopt multiple-choice evaluation or provide only coarse temporal supervision, such as single timestamps. In contrast, EG-VQA focuses on open-ended reasoning with fine-grained evidence grounding, requiring models not only to answer questions correctly but also to localize the supporting temporal evidence.

Compared with existing open-ended VideoQA benchmarks, EG-VQA places stronger emphasis on reasoning faithfulness and interpretability through explicit evidence supervision. This design enables evaluation beyond answer accuracy and supports research on evidence-grounded video reasoning.

B EG-F1: Complete Formulation

This section expands the EG-F1 metric introduced in Section 3.3.

Table 4: Comparison of existing VideoQA benchmarks and EG-VQA.

Dataset (test)	Domain	Annotation	# QAPairs	Avg. Len. (s)	Q Type	Evidence Grounding
MovieQA [Tapaswi et al., 2016]	Movie	Manual	3,138	211.4	MC	Simple Timestamp
TGIF-QA [Jang et al., 2017]	GIF	Auto/Manual	25,751	unknown	MC & OE	×
TVQA [Lei et al., 2018]	Movie	Manual	15,253	60	MC	Simple Timestamp
DramaQA [Choi et al., 2021]	Drama	Manual	3,453	3.8/93.0	MC	×
LvBench [Zhang et al., 2025]	Movie	Manual	4,155	unknown	MC	Simple Timestamp
CinePile [Rawal et al., 2024]	Movie	Auto	4,941	unknown	MC	×
MSRVTT-QA [Xu et al., 2017]	Open	Auto	72,821	15	OE	×
Pitt [Hussain et al., 2017]	Ads	Manual+Auto	unknown	unknown	OE	×
How2QA [Li et al., 2020]	Open	Manual	4,400	60	OE	Simple Timestamp
ActivityNet-QA [Yu et al., 2019]	Open	Manual	8,000	180	OE	×
VideoBench [Ning et al., 2025]	Open	Manual	4,000	56	MC	Simple Timestamp
EgoSchema [Mangalam et al., 2023]	Egocentric	Manual	5,031	180	MC	Simple Timestamp
MVBench [Li et al., 2024]	Open	Auto	4,000	15	MC	Simple Timestamp
AdsQA [Long et al., 2025]	Ads	Manual+Auto	7,859	52.9	OE	×
EG-VQA (Ours)	Open	Manual+Auto	2,889	177.6	OE	Fine-grained Timestamp

B.1 Computation Steps

Step 1: Pairwise similarity. For a ground-truth item $g_i = (t_i, d_i)$ and a prediction $p_j = (t_j, d_j)$, we compute both temporal overlap and semantic similarity:

$$\text{IoU}_{ij} = \frac{|t_i \cap t_j|}{|t_i \cup t_j|}, \quad (6)$$

$$\text{Sim}_{ij} = \cos(\phi(d_i), \phi(d_j)), \quad (7)$$

where $\phi(\cdot)$ denotes a sentence embedding function used to encode textual evidence descriptions.

Step 2: Edge construction. Valid candidate edges are defined by

$$\mathcal{E} = \{(i, j) \mid \text{IoU}_{ij} \geq \alpha \wedge \text{Sim}_{ij} \geq \beta\}, \quad (8)$$

where α and β denote the temporal IoU threshold and semantic similarity threshold, respectively. Different threshold settings are used in our experiments to evaluate grounding performance under varying levels of strictness.

Step 3: Optimal matching. A bipartite graph with edge weights

$$s_{ij} = \begin{cases} \text{IoU}_{ij} \cdot \text{Sim}_{ij}, & (i, j) \in \mathcal{E}, \\ 0, & \text{otherwise}, \end{cases} \quad (9)$$

is constructed between $\mathcal{G}$ and $\mathcal{P}$. The Hungarian algorithm is applied to obtain a maximum-weight one-to-one matching $\mathcal{M}$. Let $M = |\{(i, j) \in \mathcal{M} \mid s_{ij} > 0\}|$.

Step 4: Metric computation. Precision, recall, and F1 are computed as

$$P = \frac{M}{|\mathcal{P}|}, \quad R = \frac{M}{|\mathcal{G}|} \quad (10)$$

$$\text{EG-F1} = \frac{2PR}{P + R}. \quad (11)$$

B.2 Soft variant for training.

To mitigate reward sparsity during reinforcement learning, a soft variant is employed by removing the hard threshold in Step 2. The pairwise weight becomes $s_{ij}^{\text{soft}} = \text{IoU}_{ij} \cdot \text{Sim}_{ij}$ for all (i, j) , and soft precision/recall are defined as

$$P_{\text{soft}} = \frac{\sum_{(i,j) \in \mathcal{M}} s_{ij}^{\text{soft}}}{|\mathcal{P}|}, \quad R_{\text{soft}} = \frac{\sum_{(i,j) \in \mathcal{M}} s_{ij}^{\text{soft}}}{|\mathcal{G}|}. \quad (12)$$

The soft variant is then

$$\text{EG-F1}_{\text{soft}} = \frac{2P_{\text{soft}}R_{\text{soft}}}{P_{\text{soft}} + R_{\text{soft}}} \quad (13)$$

, which is combined with the hard EG-F1 as described in the main text.

B.3 Complexity.

Computing all pairwise similarities costs $O(|\mathcal{G}||\mathcal{P}|)$; the Hungarian algorithm adds $O(\max(|\mathcal{G}|, |\mathcal{P}|)^3)$. In practice, $|\mathcal{G}|$ and $|\mathcal{P}|$ are small (average 2.27), making the metric efficient to compute at both training and inference time.

C EG-Reasoner: Supplementary Details

C.1 Answer Reward Details

The answer reward evaluates whether the generated answer is semantically correct and consistent with the reference answer while avoiding hallucinated or unsupported content.

Evaluation Principles. We adopt a rule-guided large language model (LLM) evaluator that assesses each generated answer based on two complementary principles:

- **Inclusion principle:** the generated answer should cover as many key elements of the reference answer as possible.
- **Exclusion principle:** the answer should not contain information that contradicts or cannot be inferred from the video metadata.

Scoring Scheme. To reduce reward sparsity, especially during early training, we employ a relaxed scoring scheme that assigns partial credit:

$$R_{\text{ans}}(y) = \begin{cases} 1, & \text{fully correct,} \\ 0.5, & \text{partially correct,} \\ 0, & \text{incorrect.} \end{cases} \quad (14)$$

This design provides denser supervision signals compared to binary evaluation, enabling more stable optimization.

Prompt-based Evaluation. The evaluation is performed using a prompt-based LLM judge, which compares the generated answer against the reference answer and the associated video metadata. The prompt explicitly instructs the evaluator to follow the inclusion and exclusion principles and to output a discrete score according to the predefined rubric.

C.2 GRPO Objective Details

We adopt Group Relative Policy Optimization (GRPO) to optimize the structured generation policy.

Given an input (V, q) , the policy π_θ samples a group of candidate responses $\{y_1, \dots, y_G\}$. Each candidate is assigned a reward R_i , and the relative advantage is computed as:

$$A_i = \frac{R_i - \mu}{\sigma}, \quad (15)$$

where μ and σ denote the mean and standard deviation of rewards within the sampled group.

Let $r_i = \frac{\pi_\theta(y_i|V, q)}{\pi_{\theta_{\text{old}}}(y_i|V, q)}$ denote the importance sampling ratio. The clipped GRPO objective is:

$$\mathcal{L}(\theta) = \mathbb{E}_{(V, q), \{y_i\} \sim \pi_{\theta_{\text{old}}}} \left[\sum_{i=1}^G \min(r_i A_i, \text{clip}(r_i, 1 - \epsilon, 1 + \epsilon) A_i) - \beta_{\text{KL}} \text{KL}(\pi_\theta \| \pi_{\text{ref}}) \right], \quad (16)$$

where $\epsilon = 0.2$ is the clipping parameter and $\beta_{\text{KL}} = 0.01$ controls the strength of KL regularization.

KL Regularization. To prevent the policy from deviating excessively from the initialization, we include a KL divergence penalty with respect to a frozen reference policy π_{ref} , computed at the sequence level.

Rationale. GRPO differs from PPO in that it eliminates the need for a separate value function by using group-level statistics (μ, σ) as a baseline. This design is particularly suitable for EG-Reasoner, where evaluating the value of complex structured outputs is itself non-trivial.

C.3 Implementation Details

The key evaluation settings for our experiments on the EG-VQA benchmark are summarized in Table 5.

Table 5: Training configuration of EG-Reasoner.

Hyperparameter	Value
Base model	Qwen2.5-VL-7B-Instruct
Trainable parameters	Language model only
Frozen parameters	Visual backbone
Epochs	1
Training frames	32
Frame sampling strategy	Uniform sampling
Global batch size	4
Learning rate	1e-6
Training steps	558
Group size G	8
$\lambda_f / \lambda_e / \lambda_a$	0.1 / 0.6 / 0.3
ϵ (clip)	0.2
β_{KL}	0.01
Sentence Embeddings	all-MiniLM-L6-v2
Number of GPUs	$4 \times \text{NVIDIA A800}$
Training time	~ 50 hours
Output format	Unified structured format in Section 4

D Experimental Details

D.1 Baseline Details

We evaluate representative Video-LLMs covering proprietary systems, general-purpose open-source models, and recent reasoning-enhanced variants.

GPT-4o [Hurst et al., 2024] is a proprietary multimodal large language model capable of processing visual and textual inputs in a unified framework. We use it as a representative commercial baseline.

Gemini-2.5-Flash [Comanici et al., 2025] is a proprietary multimodal model from the Gemini family, designed to balance efficiency and reasoning capability. We include it as another commercial baseline.

Qwen2.5-VL-7B [Bai et al., 2025b] is an open-source multimodal model built upon the Qwen2.5 backbone. It supports image and video inputs and is used as a representative general-purpose open-source baseline.

InternVL-3-8B [Zhu et al., 2025]. is an open-source multimodal model from the InternVL family, designed for multimodal perception and reasoning. It is included as a general multimodal understanding baseline.

Time-R1-7B [Wang et al., 2025b] is a reasoning-oriented video model designed to improve temporal reasoning and event-level understanding. We include it to evaluate temporal reasoning capability.

TW-GRPO [Dang et al., 2025] is a GRPO-based reasoning model that introduces token-wise weighting and soft reward mechanisms to provide more fine-grained supervision. It is included as a reinforcement learning-based reasoning baseline.

VideoChat-R1-7B [Li et al., 2025] is a video reasoning model trained with reinforcement learning to enhance spatio-temporal perception and reasoning. It serves as a strong open-source video reasoning baseline.

VideoChat-R1-Thinking-7B [Li et al., 2025] extends the VideoChat-R1 framework by incorporating explicit thinking-style reasoning, and is included as a reasoning-oriented video baseline.

VideoChat-R1.5-7B [Yan et al., 2025] is an improved version of VideoChat-R1 that emphasizes test-time perceptual refinement. We include it as a recent video reasoning baseline.

D.2 Answer Evaluation Protocol

We evaluate answer correctness using an LLM-based protocol, as EG-VQA is formulated as open-ended question answering where semantically equivalent answers may have different surface forms, making traditional string-based metrics insufficient.

Given a test set of N samples, let x_i denote the model response and y_i the reference answer. The strict accuracy is defined as:

$$\text{Acc}_{\text{strict}} = \frac{1}{N} \sum_{i=1}^N \text{Match}_{\text{full}}(x_i, y_i), \quad (17)$$

where $\text{Match}_{\text{full}}(\cdot)$ returns 1 if the response fully matches the reference answer in meaning.

To account for partially correct answers, we define relaxed accuracy as:

$$\text{Acc}_{\text{relaxed}} = \frac{1}{N} \sum_{i=1}^N (\text{Match}_{\text{full}}(x_i, y_i) + \gamma \cdot \text{Match}_{\text{partial}}(x_i, y_i)), \quad (18)$$

where $\text{Match}_{\text{partial}}(\cdot)$ indicates partial correctness and γ controls the credit assigned to partially correct answers. In our experiments, $\gamma = 0.5$.

We use Gemini-2.5-Pro as the evaluator, which assesses semantic correctness based on the question, reference answer, and model response. The same evaluation protocol is applied to all baselines and EG-Reasoner. We additionally manually inspect a subset of evaluation results and observe high consistency between the LLM judge and human judgment, suggesting that the automatic evaluation protocol is reasonably reliable for open-ended VideoQA evaluation.

D.3 Evidence Evaluation Settings

Table 6: Evaluation configuration used for EG-VQA.

Evaluation setting	Value
Benchmark	EG-VQA
Task format	Open-ended QA
Evaluation frames	64
Frame sampling strategy	Uniform sampling
Maximum visual resolution budget	$256 \times 28 \times 28$
Output format	Unified structured answer-and-evidence format
Answer metric	Strict / relaxed accuracy
Evidence metric	EG-F1 and event-level F1

We evaluate evidence grounding using both event-level temporal F1 and the proposed EG-F1 metric.

For temporal localization, we report F1 scores under multiple IoU thresholds $\tau \in \{0.1, 0.3, 0.5, 0.7\}$ to evaluate localization performance at different granularities.

For EG-F1, we evaluate multiple threshold combinations (α, β) , where α denotes the temporal IoU threshold and β denotes the semantic similarity threshold. Specifically, we report results under $(0.3, 0.5)$, $(0.3, 0.75)$, and $(0.5, 0.75)$, corresponding to increasingly strict grounding criteria.

D.4 Evaluation Configuration

The key evaluation settings for our experiments on the EG-VQA benchmark are summarized in Table 6.

E Additional Case Studies

We provide additional qualitative examples to further analyze the behavior of EG-Reasoner on evidence-grounded video reasoning.

Input Video



Question: What are the steps the woman takes to prepare for using the bike?

GT Evidence: 00:04 00:48 00:48 01:52
bike visuals are shown; a woman adjusts the seat. She wears a heart monitor and rides the connected bike.

GT Answer: She adjusts the bike seat and puts on a heart monitor before using the bike.

 **EG-Reasoner (Ours)**

<evidence>
 Time:00:00-00:54, Des: The bike is displayed from different angles, followed by the woman adjusting its seat height. ✓
 Time:01:50-01:50, Des: She attaches a heart rate monitor to herself and begins riding the connected exercise bike. ✓
</evidence>

<think> The video shows the woman first setting up the bike by changing the seat position. After that, she wears a heart monitor and proceeds to use the bike. These two actions — adjusting the seat and putting on the monitor — happen sequentially before she starts cycling.
</think>

<answer>
 She prepares by first adjusting the bike seat, then putting on a heart monitor before beginning to ride. ✓
</answer>

Figure 8: Qualitative example of successful evidence-grounded reasoning. EG-Reasoner correctly identifies both the preparation steps and the corresponding temporal evidence segments.

Input Video



Question: What steps are involved in preparing the cereal box before designing it?

GT Evidence: 00:41 01:18 01:28 02:05
trim excess cereal box parts. glue the box closed..
 01:18 01:28 02:05 02:46
match the small box size. cover all sides with cardboard.

GT Answer: Cut the excess parts, match size, close with glue, paste cardboard..

 **EG-Reasoner (Ours)**

<evidence>
 Time:00:10-02:59, Des: Cutting paper to make a cereal box. ✗
 Time:03:00-03:52, Des: Assembling the cereal box from cut paper. ✗
</evidence>

<think> The video segments I identified show that from 00:10 to 02:59, the action is cutting paper to make a cereal box, and from 03:00 to 03:52, it is assembling the cereal box from cut paper. Based on my evidence, I conclude that the answer is cutting the paper and assembling the box.
</think>

<answer>
 The steps involve cutting the paper and assembling the box. !
</answer>

Figure 9: Qualitative example of incomplete procedural grounding. EG-Reasoner captures the overall procedure but misses several fine-grained intermediate steps required for detailed reasoning.

F Prompt templates

Figures 10–16 present the complete set of prompt templates designed in this paper. Specifically, Fig. 10 illustrates the prompt template for descriptive question generation. Figs. 11, 12, and 13 show the prompts for temporal, causal, and counterfactual reasoning generation, respectively. Fig. 14 presents the prompt used for model cross-validation. Fig. 15 provides the training and inference prompt for all MLLMs. Fig. 16 gives the prompt for judge model evaluation.

You are an **expert video understanding and descriptive question designer**. Your task is to generate **high-quality descriptive open-ended questions** that require describing **what is observable, stated, or directly inferable** from timestamped segments of a video.

Input Format:

You will receive a JSON object containing:

- `title`: overall content focus of the video
- `timestamps`: a list of `[start, end]` time segments
- `descriptions`: a list of corresponding descriptions for each segment

Task Instructions:

1. Generate **1 to 3 open-ended questions**, adjusting the number based on the richness of video content available. Each question must be **descriptive** in nature, asking about **what** is observable, stated, or directly inferable from the descriptions. Focus on: events, actions, objects, people, states, locations, relationships, and summaries. If the video content does not support genuine descriptive questions, output `{"questions": []}` with no additional text.
2. Questions may require integrating information from **one or more timestamp segments**, depending on the descriptive content being tested.
3. Questions must be **self-contained** and clear. **Do not use any words that explicitly or implicitly reference the video's timestamps or segment order.**
4. Base your questions **strictly on the provided descriptions**. Do not introduce outside knowledge, assumptions, or speculative content (e.g., the person's identity, emotions, intentions, or behind-the-scenes details).
5. The question should be **concise, and low in information density**. Do **not** ask temporal reasoning questions that rely on order/sequence (e.g., "what happened after/before/first/next/during", "what is the sequence of"). Questions should **not** test temporal order.

Output Format:

Return a valid JSON array. The array can contain **multiple items** (1 to 3 as appropriate).

Each item follows the structure below; the evidence array can also contain multiple segment entries:

```
[
  {
    "question": "A clear, open-ended question string testing descriptive reasoning.",
    "reasoning": "A step-by-step explanation (2-4 sentences) showing how evidence supports the answer.",
    "answer": "A concise answer no more than 30 words.",
    "evidence": [
      { "timestamp": [start, end], "description": "string" }
    ]
  }
]
```

Constraints:

- The `timestamp` and `description` fields within each evidence item must be copied **verbatim** from the input, without modification, omission, or addition.
- Evidence may come from **one or more** segments; when multiple segments are used, list them in chronological order.

Now generate descriptive questions for the following video data: {metadata}

Figure 10: Prompt template for descriptive question generation.

You are an **expert video understanding and temporal reasoning question designer**. Your task is to generate **high-quality temporal reasoning open-ended questions** that require understanding **order/sequence, change detection, and process/procedure** across timestamped segments from a video.

Input Format:

You will receive a JSON object containing:

- `title`: overall content focus of the video
- `timestamps`: a list of `[start, end]` time segments
- `descriptions`: a list of corresponding descriptions for each segment

Task Instructions:

1. Generate **1 to 3 open-ended questions**, adjusting the number based on the richness of temporal relationships available. If the video content does not support genuine temporal reasoning, output `{"questions": []}` with no additional text. Each question must be **temporal reasoning** in nature, testing one of the following:

- **Order/Sequence**: Understanding what happens before/after/during specific events (ask "what happens before/after", "what occurs first/last")
- **Change Detection**: Recognizing how something changes between different points in time (ask "how does X change", "what difference appears")
- **Process/Procedure**: Identifying the sequence of steps in a process or workflow (ask "what are the steps", "in what order")

2. Questions must be **self-contained** and clear. **Do not use any words that explicitly or implicitly reference the video's timestamps or segment order.**

3. Base your questions **strictly on the provided descriptions**. Do not introduce outside knowledge, assumptions, or speculative content (e.g., the person's identity, emotions, intentions, or behind-the-scenes details).

4. The question should be **concise, and low in information density**, shouldn't contain information that allows guessing the answer **without watching the video**.

5. **Ensure diversity**: Do not generate duplicate questions that ask the same thing in different wording. Each question should explore a different temporal relationship in the video.

Output Format:

Return a valid JSON array. The array can contain **multiple items** (1 to 3 as appropriate).

Each item follows the structure below; the evidence array can also contain multiple segment entries:

```
[
  {
    "question": "A clear, open-ended question string testing descriptive reasoning.",
    "reasoning": "A step-by-step explanation (2-4 sentences) showing how evidence supports the answer.",
    "answer": "A concise answer no more than 30 words.",
    "evidence": [
      { "timestamp": [start, end], "description": "string" }
    ]
  }
]
```

Constraints:

- The `timestamp` and `sentence` fields within each evidence item must be copied **verbatim** from the input, without modification, omission, or addition.
- Evidence may come from **one or more** segments; when multiple segments are used, list them in chronological order.

Now generate temporal reasoning questions for the following video data: {metadata}

Figure 11: Prompt template for temporal question generation.

You are an **expert video understanding and causal reasoning question designer**. Your task is to generate **high-quality causal reasoning open-ended questions** that require identifying and explaining cause-effect relationships based on factual evidence from timestamped segments of a video.

Input Format:

You will receive a JSON object containing:

- `title`: overall content focus of the video
- `timestamps`: a list of `[start, end]` time segments
- `descriptions`: a list of corresponding descriptions for each segment

Task Instructions:

1. Generate **1 to 3 open-ended questions**, adjusting the number based on the richness of causal connections available. Each question must be **causal** in nature, asking **"why"** or **"how"** something happened based on evidence from the video. If the video content does not support genuine multi-segment reasoning or lacks causal chains, output `{"questions": []}` with no additional text rather than forcing questionable ones.
2. Each question **must**:
 - Require integrating information from **at least two distinct timestamp segments** (non-adjacent segments are encouraged).
 - Be framed as either:
 - **Cause-seeking**: "Why did X happen?", "How did X happen?" or "What caused X?" (X is an outcome)
 - **Effect-seeking**: "What happened as a result of Y?" or "What effect did Y have?" (Y is an action/event)
 - Be **self-contained** and clear. **Do not** use any words that explicitly or implicitly reference the video's timestamps or segment order.
 - **Not** include the answer or obvious causal cues in the question itself. The question should name only the event/outcome of interest, without explaining its cause or consequence.
 - Require the answer to be logically deducible **only** from the causal evidence explicitly shown or described in the video segments.
3. Base your questions **strictly on the provided descriptions**. Do not introduce outside knowledge, assumptions, or speculative content (e.g., the person's identity, emotions, intentions, or behind-the-scenes details). Do not invent any fact, event, or detail not explicitly stated or logically implied.
4. The question should be **concise, abstract and low in information density**:
 - Use generic terms (e.g., "the person", "the individual", "the object") rather than specific labels (e.g., "the gymnast", "the chef") that reveal professional or situational context.
 - Do **not** mention unobserved entities or consequences (e.g., "audience", "judges", "reaction") in the question, as this leaks answer cues.
 - Should **not** contain specific action descriptions or visible states that would give away the answer (e.g., avoid phrases like "using the rope," "bending the knee," or "pulling the row"). Instead, phrase it in terms of the **outcome, role, or purpose** that requires viewing the video to understand. The question should be answerable **only** by synthesizing evidence from multiple segments.
 - **Critical**:
 - Do **not** state the cause when asking about an effect (e.g., avoid "Why did the tree look beautiful after adding lights?" — this gives away the cause).
 - Do **not** state the effect when asking about a cause (e.g., avoid "What did marinating the chicken cause to happen to the final dish?" — this gives away the effect domain).
 - Do **not** include any phrasing that situates the event within a larger sequence as a positional marker (e.g., "before serving", "during the routine", "after adding X", "when making Z"). The question should ask about the event itself.
 - Do **not** phrase the question as a full action that includes the surrounding context. Instead, use a minimal or noun-phrase structure. The question should be short and abstract, requiring the video to reveal what that element is used for and what consequence follows.
5. **Ensure diversity**: Do not generate duplicate questions that ask the same thing in different wording. Each question should explore a different causal relationship in the video.

Output Format:

Return a valid JSON array. The array can contain **multiple items** (1 to 3 as appropriate).

Each item follows the structure below; the evidence array can also contain multiple segment entries:

```
[
  {
    "question": "A clear, open-ended question string testing descriptive reasoning.",
```

```

    "reasoning": "A step-by-step explanation (2-4 sentences) showing how evidence supports the answer.",
    "answer": "A concise answer no more than 30 words.",
    "evidence": [
      { "timestamp": [start, end], "description": "string" }
    ]
  }
]

```

Constraints:

- The `timestamp`` and `description`` fields within each evidence item must be copied **verbatim** from the input.
- Evidence may come from **one or more** segments; when multiple segments are used, list them in chronological order.
- **Critical:** The `question`` field should be as short and abstract as possible. Avoid describing the surrounding scene, the broader task, or the role of the action/event. The video should be required to understand both the context and the causal chain.

Now generate causal reasoning questions for the following video data: {metadata}

Figure 12: Prompt template for causal question generation.

You are an **expert video understanding and counterfactual reasoning question designer**. Your task is to generate **high-quality counterfactual reasoning open-ended questions** that require reasoning about hypothetical scenarios ("what if") based on factual evidence from timestamped segments of a video.

Input Format:

You will receive a JSON object containing:

- `title`: overall content focus of the video
- `timestamps`: a list of `[start, end]` time segments
- `descriptions`: a list of corresponding descriptions for each segment

Task Instructions:

1. **First, evaluate the video content:** Determine if it contains clear causal relationships, dependencies, or alternative possibilities that support meaningful counterfactual reasoning.
 - **Unsuitable scenarios** (output `{"questions": []}`): Static scenes, purely descriptive monologues without actions, videos lacking sequential logic, or scenarios where altering a step leads to trivial outcomes.
 - If suitable, generate **1 to 3 open-ended questions**, adjusting the number based on the richness of causal connections available. Prefer quality over quantity.
2. **Ensure diversity:** Do not generate duplicate questions. Each question should test a distinct counterfactual scenario or explore a different causal relationship.
3. **Ensure reasoning complexity:** Prioritize questions that require synthesizing information from **at least two distinct timestamped segments** to establish a causal chain.
4. Each question must:
 - Begin with a counterfactual premise ("What if...", "If... not/had...")
 - Alter **one specific event or condition** from the actual video, while keeping all other events unchanged
 - Do **not** include in the question any description of the consequence, downstream step, or final context (e.g., do not say "before adding to the salad", "before serving", "during the routine", "when making the dish"). The question should name only the altered event itself, without situating it within the broader sequence or outcome. The causal relationship must be inferred from the video, not stated in the question.
 - Do **not** include in the question any description that situates the altered event within a larger action or process. The question should name only the core element being altered (e.g., an object, a substance, a tool, or a simple action), not the full action that surrounds it. Avoid phrasing like "decorated with lights" or "placed the star on the tree"; instead use structures like "lights were not used" or "the golden star was not used".
 - Require the answer to be logically deducible from: (a) the altered premise, and (b) the actual events/relationships shown in the video
5. Base your questions **strictly on the provided descriptions**. Do not invent any fact, event, or detail not explicitly stated or logically implied.

Output Format:

Return a valid JSON array. The array can contain **multiple items** (1 to 3 as appropriate).

Each item follows the structure below; the evidence array can also contain multiple segment entries:

```
[
  {
    "question": "A clear, open-ended question string testing descriptive reasoning.",
    "reasoning": "A step-by-step explanation (2-4 sentences) showing how evidence supports the answer.",
    "answer": "A concise answer no more than 30 words.",
    "evidence": [
      { "timestamp": [start, end], "description": "string" }
    ]
  }
]
```

Constraints:

- The `timestamp` and `description` fields within each evidence item must be copied **verbatim** from the input.
- Evidence may come from **one or more** segments; when multiple segments are used, list them in chronological order.
- **Critical:** The `question` field should be as short and abstract as possible. Avoid describing the surrounding scene, the broader task, or the role of the element being altered. The video should be required to understand both the context and the causal chain.

Now generate counterfactual reasoning questions for the following video data: {metadata}

Figure 13: Prompt template for counterfactual question generation.

You are a strict but fair **video question-answer validator**. Your task is to verify whether a generated question and its answer are reasonable based on the provided video metadata.

Input Format:

- ``metadata``: Contains video duration, timestamp segments, and corresponding descriptions for each segment
- ``question``: The generated question to evaluate
- ``question_type``: Type of question, which belongs to one of the following four types: causal, counterfactual, temporal, descriptive
- ``generated_answer``: The generated answer to evaluate

Validation Criteria:

1. **Question Reasonableness (0-10 points)**
 - Does the question require watching the video to answer? (Cannot be answered by common sense alone)
 - Is the question relevant to the video content described in metadata?
 - Is the question clear, specific, and answerable based on the metadata?
2. **Answer Correctness (0-10 points)**
 - Is the answer directly supported by the metadata? (No hallucination)
 - **REASONABLE INFERENCES ARE ENCOURAGED**: If the answer makes minimal, logical inferences that are directly implied by the metadata (e.g., "to remove moisture" when metadata says "wipe with dry cloth"), this is acceptable.
 - Does the answer **DIRECTLY** address the specific question being asked? Not just being relevant or true, but actually answering what was asked (Not off-topic)
 - Is the answer concise and accurate based on available information?
3. **Metadata-Answer Alignment (0-10 points)**
 - Can all claims in the answer be traced back to specific segments?
 - Are there any contradictions between answer and metadata?

Scoring Guide:

- Total Score ≥ 25 : PASS
- Total Score < 25 : FAIL
- **Automatic FAIL only if these critical issues exist:**
 - * Answer contains information that directly contradicts metadata
 - * Answer invents major events/objects/actions not present in metadata at all
 - * Question is completely unrelated to the video content
 - * Question can be answered without any reference to the video (pure common sense)

Now validate the following:

Video metadata: {metadata}

Question: "{question}"

Question Type: {question_type}

Generated Answer: "{answer}"

Then, provide your final judgment in **valid JSON format only** (no additional text before or after):

```
{{
  "Score": <total_score>,
  "Pass": <true_or_false>
}}
```

Figure 14: Prompt template for model cross-validation.

You are a video understanding assistant. Given a video and a question, respond in the following format: First output the seen relevant video segments within <evidence> </evidence> tags in chronological order. Then based on the observed video and evidence, analyze and reason through the question. The reasoning process MUST BE enclosed within <think> </think> tags. The final answer MUST BE put in <answer> </answer> tags.

Format: <evidence>Time:MM:SS-MM:SS, Des: description (one or more lines)</evidence>. <think> reasoning based on evidence </think>. <answer> concise direct answer </answer>.

IMPORTANT RULES:

- Output EXACTLY 3 sections: <evidence>, <think>, <answer>
- Evidence: Each line MUST follow: "Time:MM:SS-MM:SS, Des: ..."
- NO repeated descriptions or time segments
- Think: 2-4 sentences maximum, based ONLY on the evidence above
- Answer: 1 short sentence, no explanations

Now answer the following question:
Question: {question}

Figure 15: Prompt template for training and inference.

You are an expert specializing in evaluating whether a respondent's answer after watching a video matches the golden answer. We will provide the video segments' descriptions, question, golden answer, and the response to be judged below.

Video Segments' Descriptions (in chronological order):
{metadata}

Question:
{question}

Golden Answer:
{golden_answer}

Response to be judged:
{model_answer}

Rules:

1. If the response to be judged contains ALL key information of the golden answer or expresses the same meaning using other sentences or synonyms, it is considered a match, and the output is 1.
2. If the response to be judged does NOT contain the key information from the golden answer, it is considered a mismatch, and the output is 0.
3. The response to be judged should NOT contain any content that is contradictory, conflicting, or unreasonable when inferred from the video content description. If such content exists, it is considered a mismatch, and the output is 0.
4. If the response to be judged contains MOST of the key information of the golden answer, and does NOT contain any information that is contradictory, conflicting, or unreasonable when inferred from the video content description, it is considered a partial match, and the output is 0.5.

Instructions:
Follow the format below and do not give any extra outputs:
Score: 0 (if the response does not match)
Score: 0.5 (if the response partially matches)
Score: 1 (if the response matches)

Figure 16: Prompt template for model evaluation.